PREDICTION OF CONCRETE STRENGTH USING UPV

Submitted to

PROF. RAJIV GUPTA

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BY

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1.Introduction..... 3
2. Methodology..... 4
2.1 Data Preprocessing..... 4
2.2 Physics-Informed Neural Network (PINNs) Model Design.....14
2.3 Loss Functions and Training Process..... 15
3. Insights and Observations..... 17
5. Deployment.....20
5. Conclusion..... 21
6. Limitations.....21

1.Introduction

The integration of Artificial Intelligence (AI) in civil engineering marks a transformative shift in how traditional engineering challenges are addressed. With advancements in computational power and algorithmic capabilities, AI has proven to be a versatile tool capable of optimizing design processes, improving construction efficiency, and enhancing the predictive accuracy of models. Among the various AI-driven approaches, Physics-Informed Neural Networks (PINNs) stand out as a powerful framework for incorporating domain knowledge into machine learning models. By embedding empirical equations and physical constraints directly into the learning process, PINNs enable a seamless blend of data-driven insights and theoretical principles.

In this study, we explore the application of PINNs for predicting concrete strength, a critical factor in construction and structural performance. Accurate predictions of concrete strength are essential for ensuring the safety, durability, and cost-effectiveness of construction projects. However, conventional methods often require extensive experimental data and may lack the flexibility to adapt to diverse conditions. By leveraging AI and domain knowledge, this project aims to address these limitations and deliver robust, physically consistent predictions.

This document provides a comprehensive overview of the methodology, insights, and results achieved using a PINNs - based model for concrete strength prediction. From preprocessing the dataset and designing the neural network architecture to implementing advanced loss functions and analyzing model performance, this work highlights the potential of AI in revolutionizing civil engineering practices.

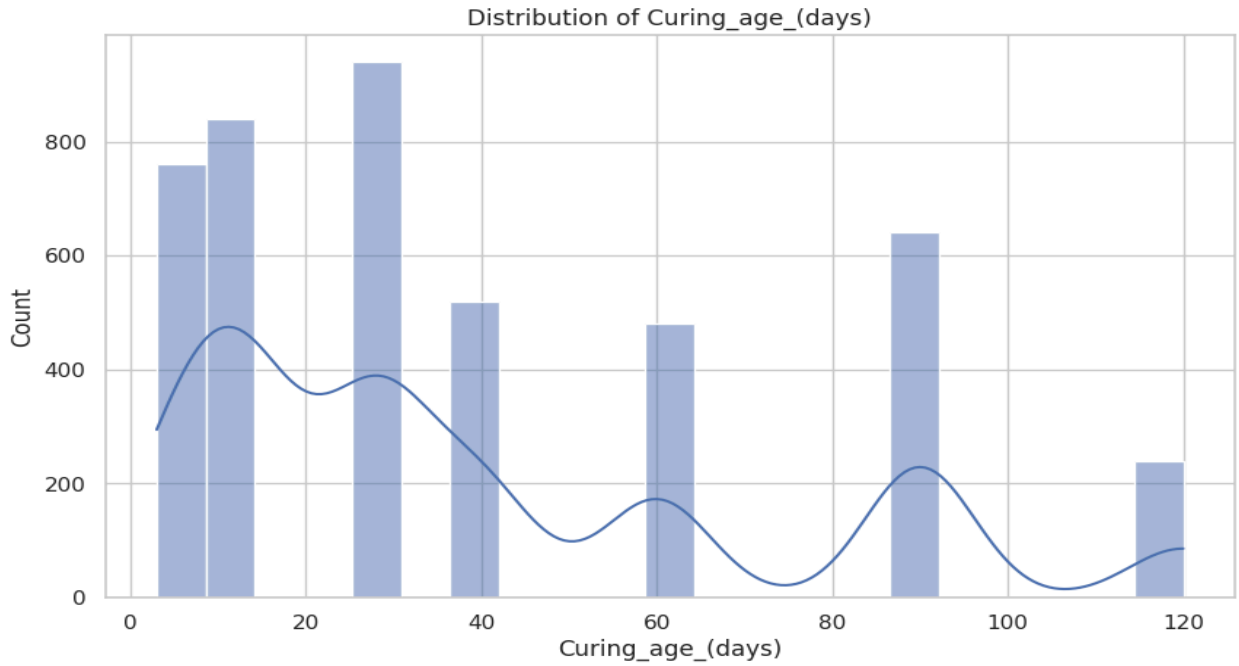
2. Methodology

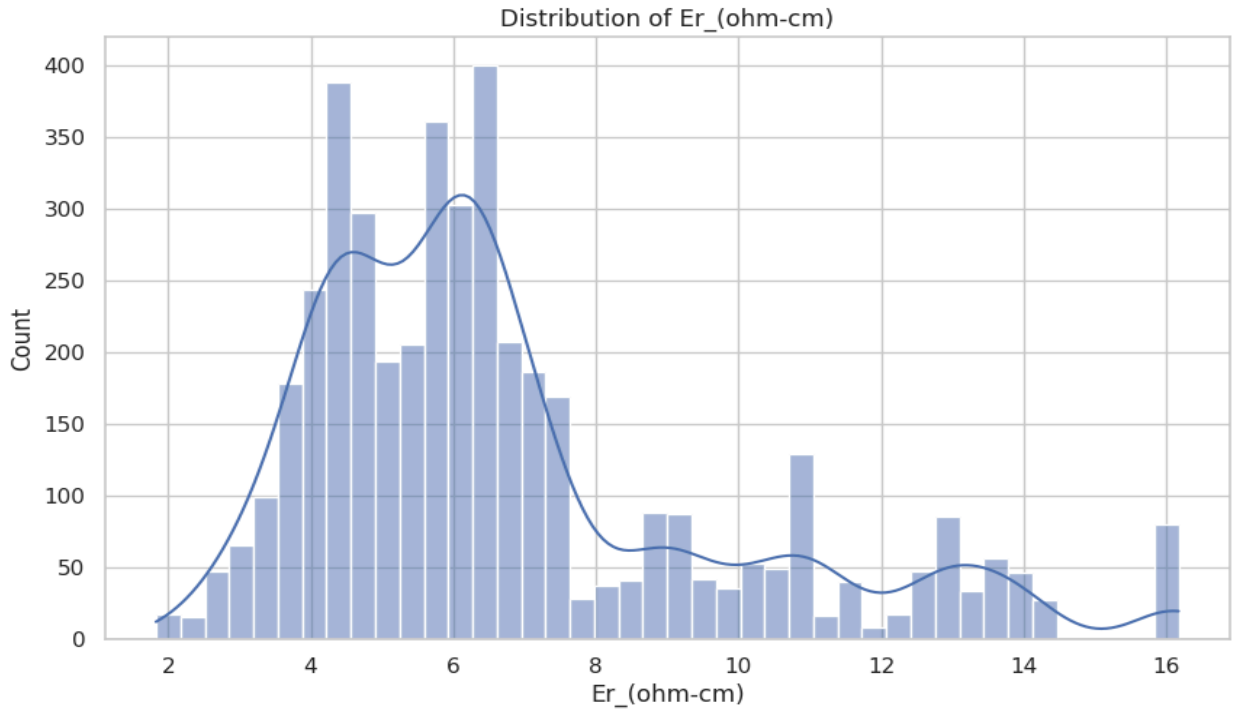
The methodology involves multiple steps to build a reliable and efficient PINN model. This includes data preprocessing, model design, loss function integration, and an effective training process. Below, each step is described in detail:

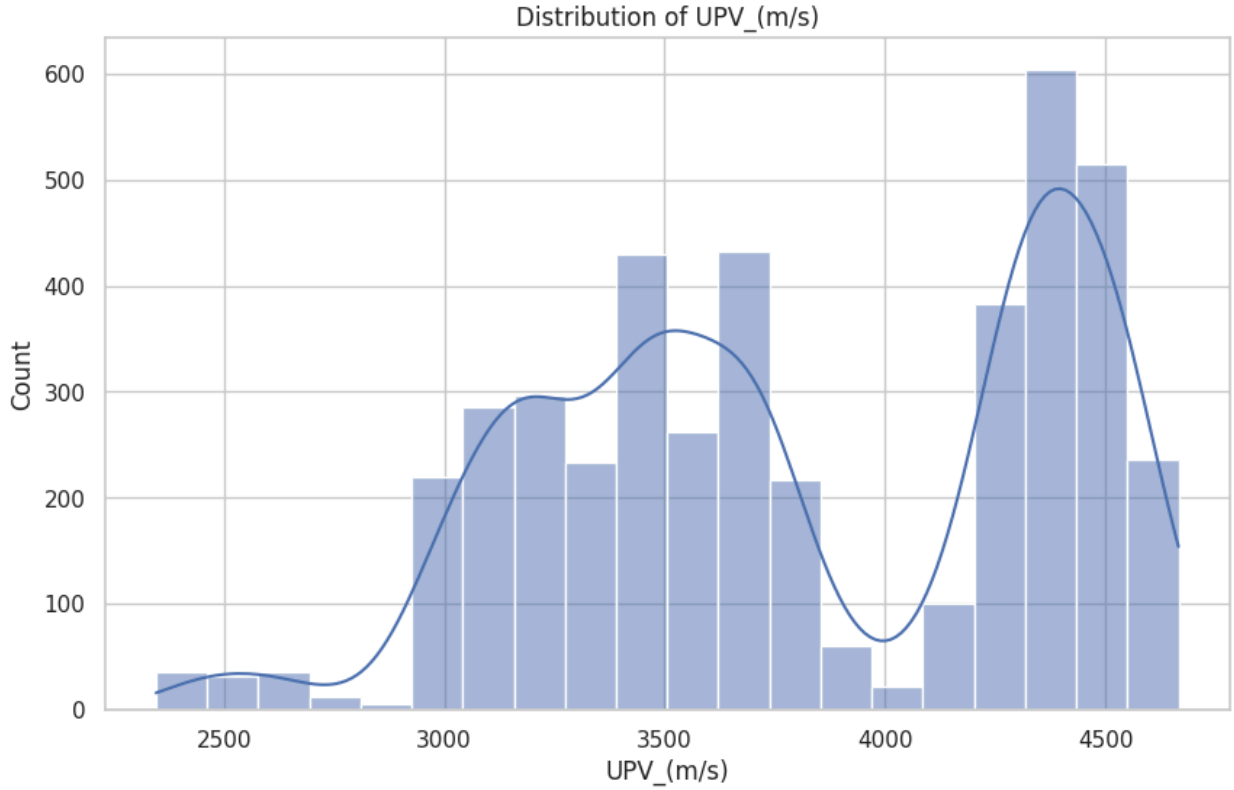
2.1 Data Preprocessing

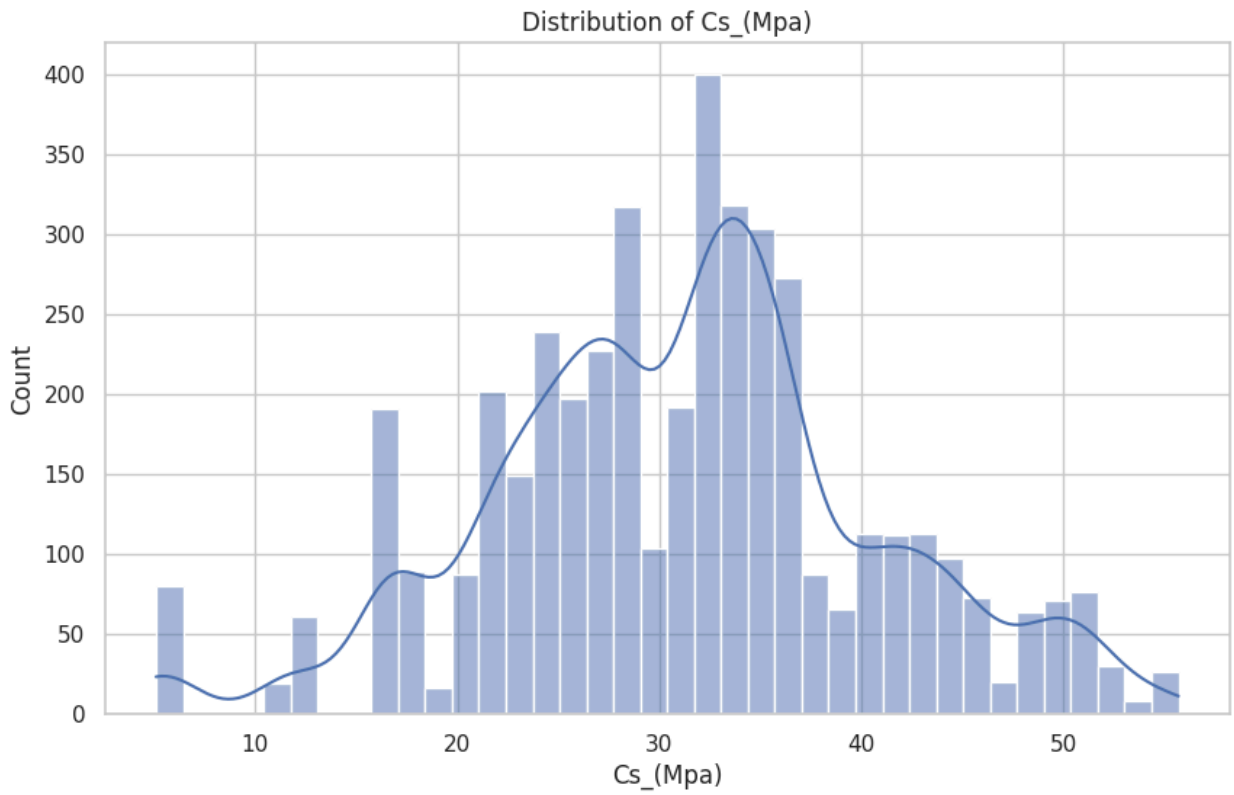
- **Dataset Description**: The dataset used for this project contains key features that influence concrete strength, including **Type of Aggregates, Curing Age (days), UPV (m/s), ER (ohm-cm), and Concrete Strength (Mpa)**. It was essential to preprocess this data to ensure high-quality inputs for model training.

1. **Distribution of different parameter value in dataset**

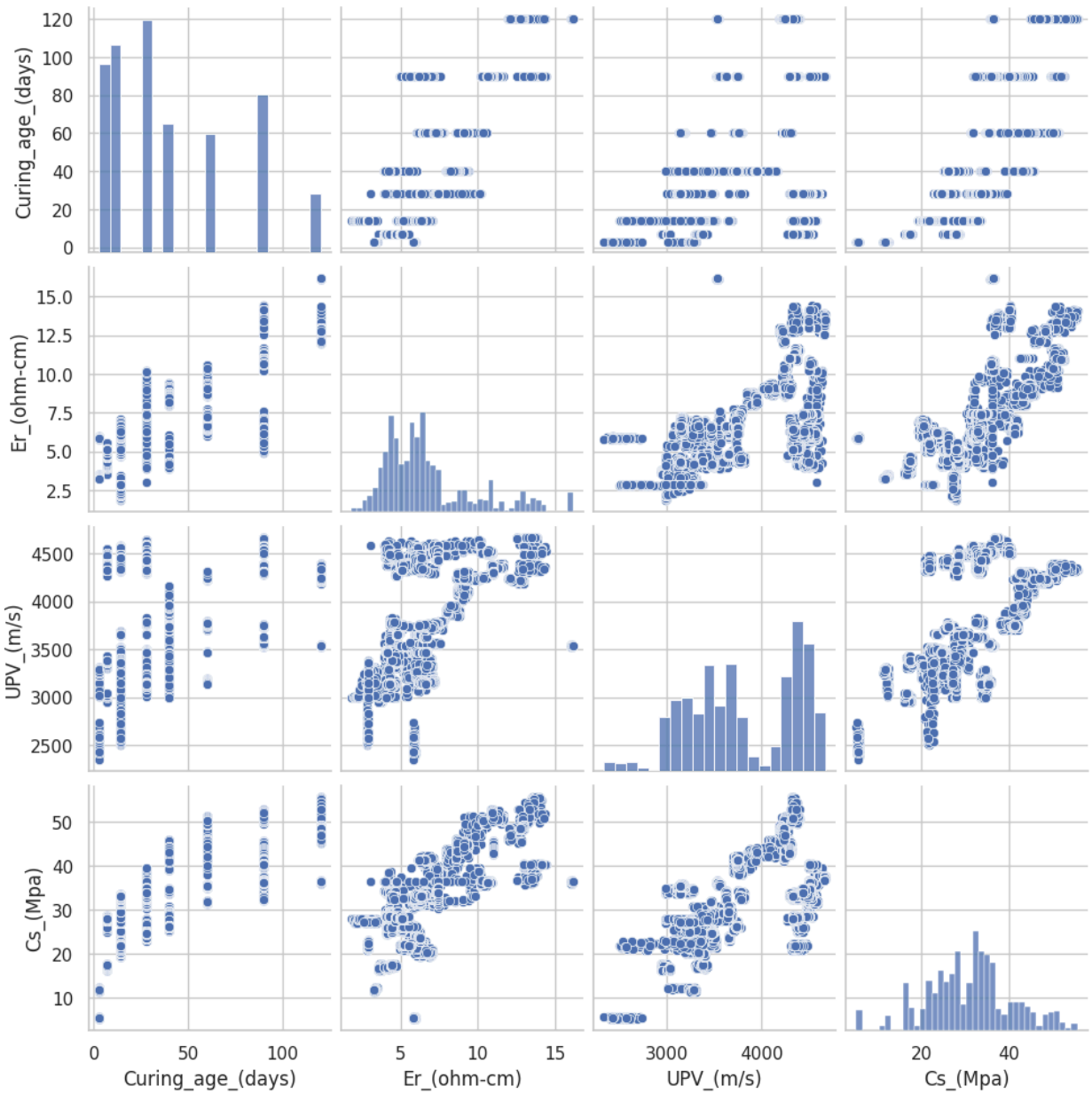



Distribution of Er_(ohm-cm)
Count
Er_(ohm-cm)
0
50
100
150
200
250
300
350
400
2
4
6
8
10
12
14
16

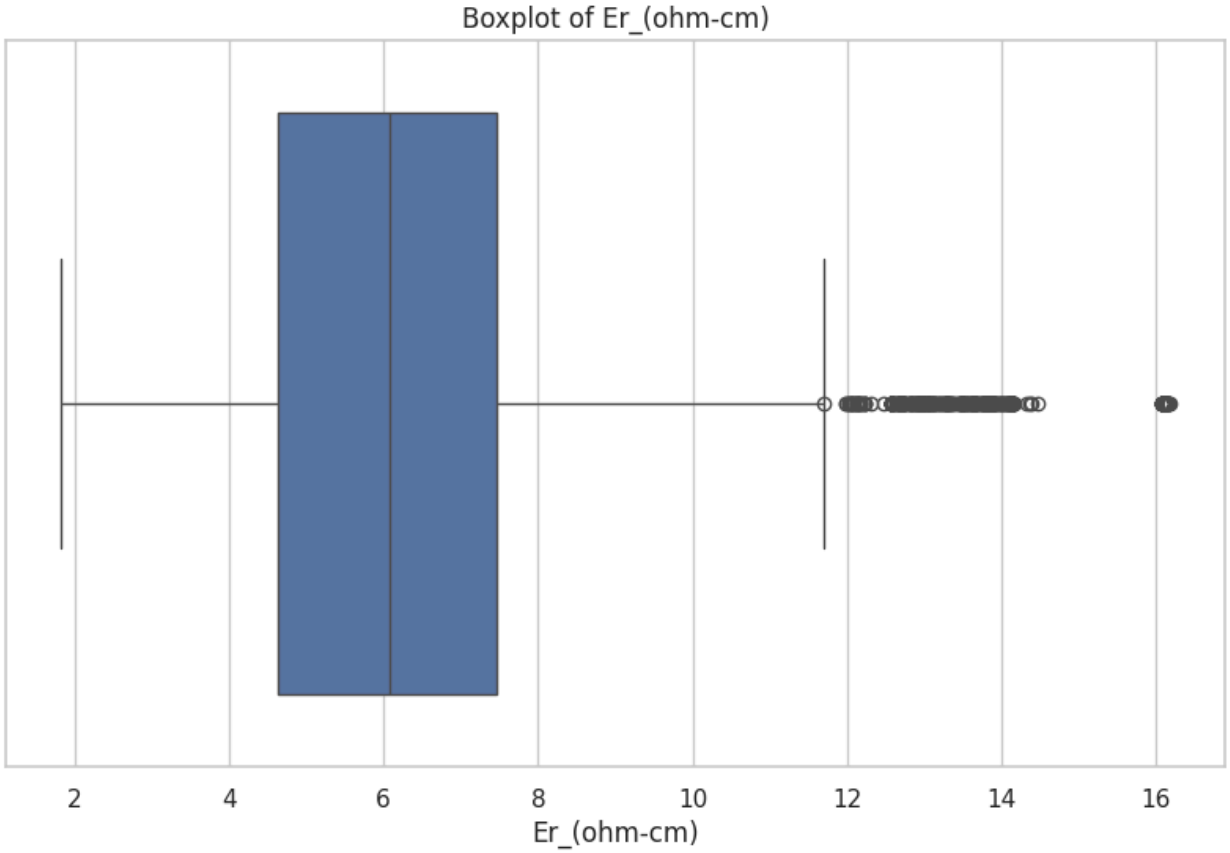

Distribution of UPV_(m/s)
Count
UPV_(m/s)
0
100
200
300
400
500
600
2500
3000
3500
4000
4500

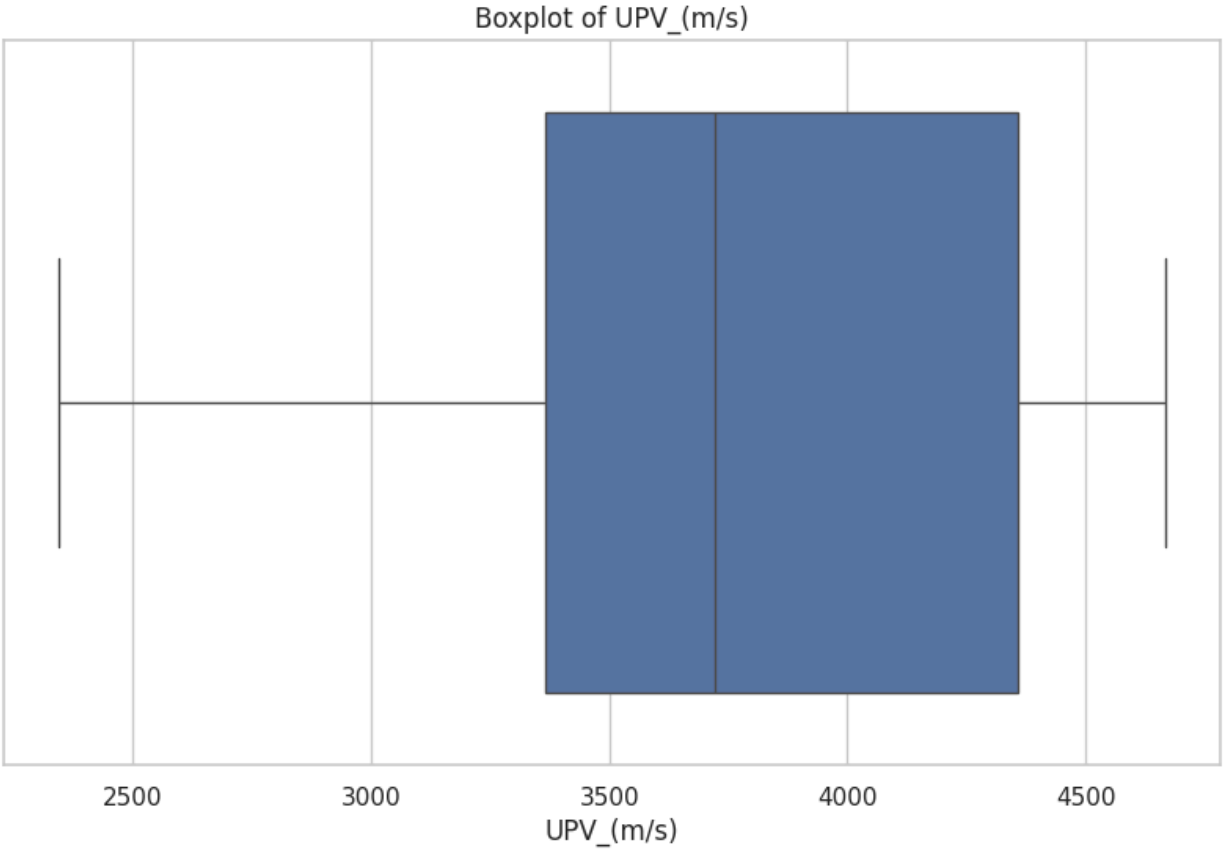

Distribution of Cs_(Mpa)
Count
Cs_(Mpa)
0
50
100
150
200
250
300
350
400
10
20
30
40
50

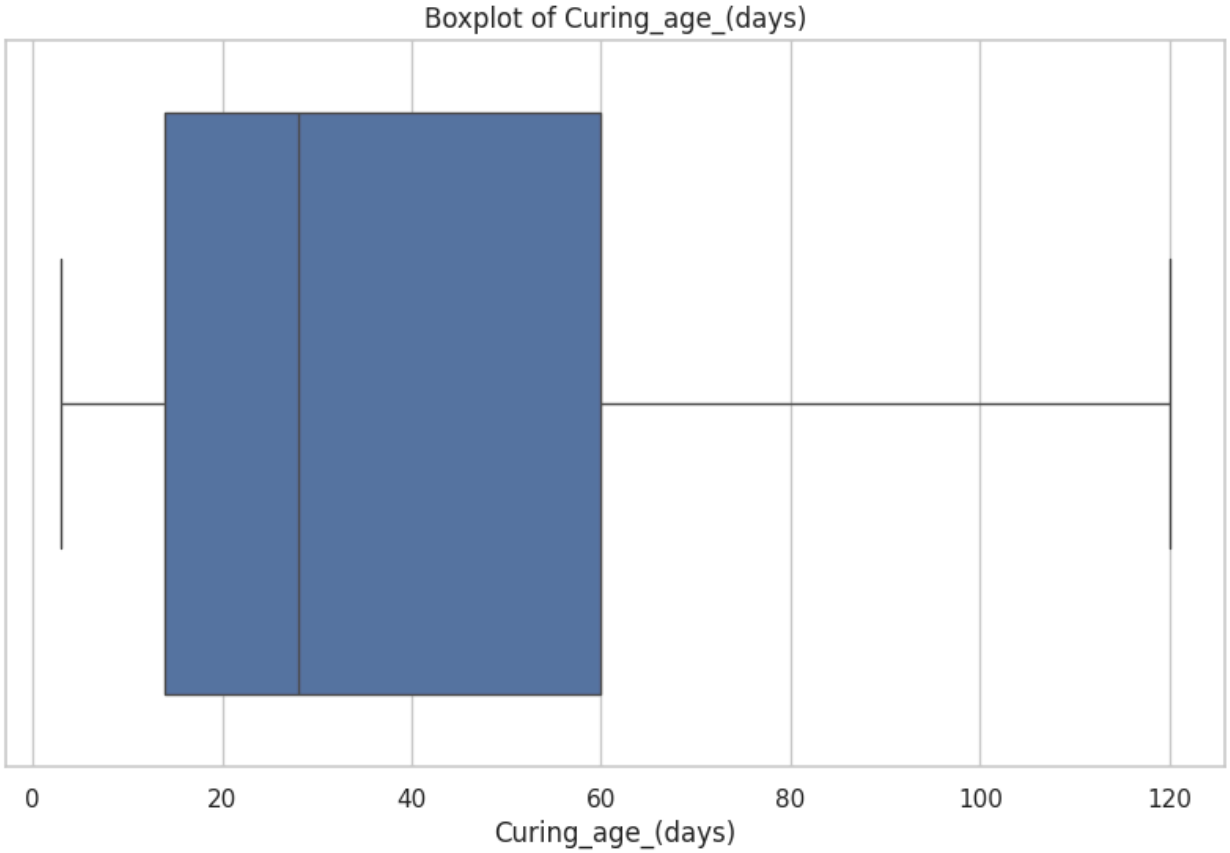
2. Pairwise Distribution Plots

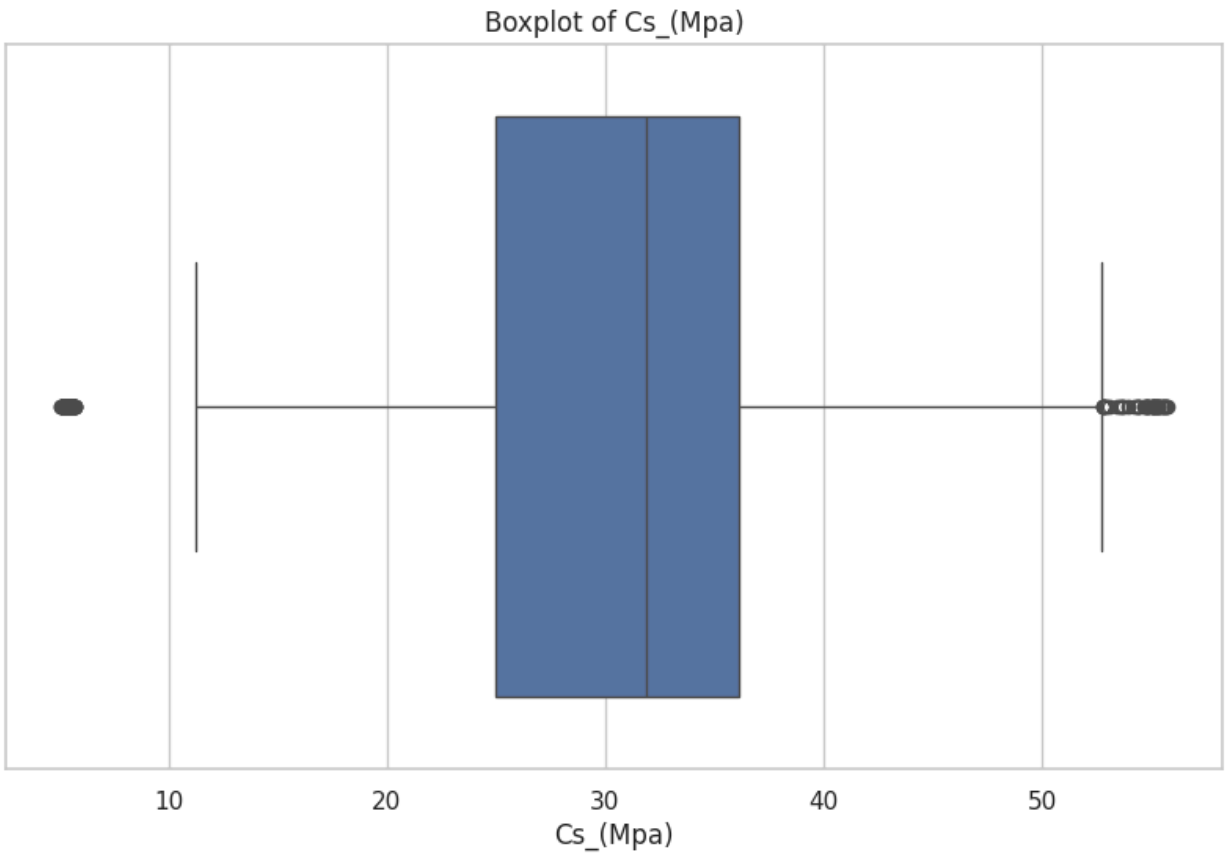


3. Box Plot

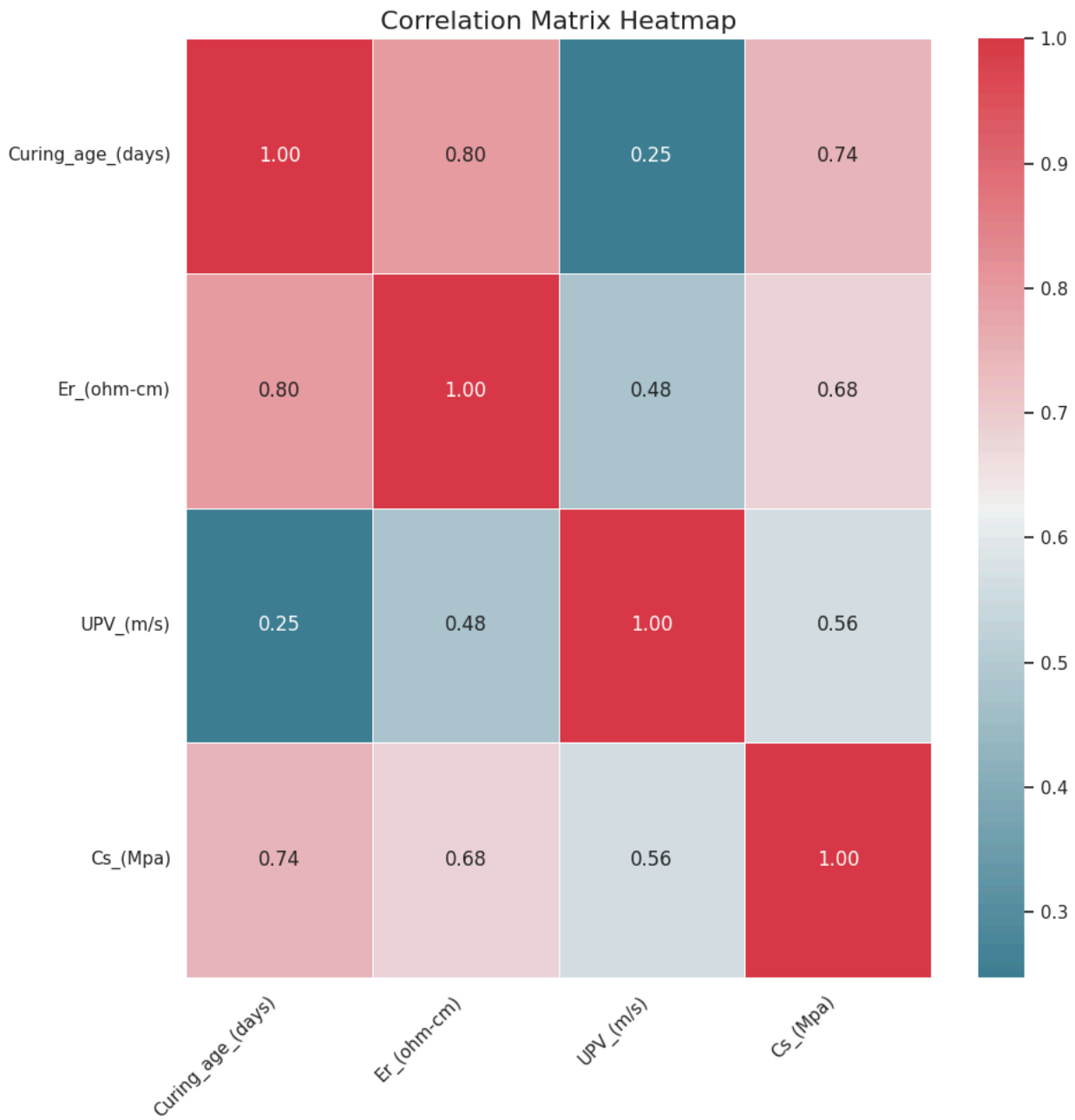

Boxplot of Er_(ohm-cm)
2
4
6
8
10
12
14
16
Er_(ohm-cm)


Boxplot of UPV_(m/s)
2500
3000
3500
4000
4500
UPV_(m/s)


Boxplot of Curing_age_(days)
0
20
40
60
80
100
120
Curing_age_(days)



4. **Correlation Matrix**



- **Feature Encoding**: The **Type of Aggregates** feature is categorical, so it was converted to numerical codes. This conversion is necessary to allow the neural network to process categorical information effectively.

```
[8] print(set(df['Type_of_aggregates']))
{'Recycled ', 'Volcanic', 'Crushed', 'Rounded'}
```

```
print(set(data['Type_of_aggregates']))
{0, 1, 2, 3}
```

- **Feature Standardization**: The numerical features, namely **Curing Age, UPV, and ER**, were standardized using **StandardScaler**. Standardization is crucial to normalize these features, ensuring that they have a mean of zero and a standard deviation of one. This helps in achieving faster convergence and stable training.

```
print((data[numerical_features]).head())
   Curing_age_(days)  UPV_(m/s)  Er_(ohm-cm)
0          -1.124258  -2.439777    -0.301336
1          -1.124258  -2.479605    -0.273830
2          -1.124258  -2.458265    -0.257198
3          -1.124258  -2.449187    -0.298070
4          -1.124258  -2.495933    -0.301336
```

- **Data Splitting**: The dataset was split into training, validation, and test sets to assess model performance. **70% of the data was used for training**, **15% for validation**, and **15% for testing**. This split helps to evaluate the model's generalizability and mitigate overfitting.

```
Dataset sizes:
Training: 3093 samples
Validation: 664 samples
Test: 663 samples
```

2.2 Physics-Informed Neural Network (PINNs) Model Design

- **Neural Network Architecture**: The PINN model is designed with a **fully connected neural network** containing **three hidden layers**, each with **128 neurons**. We used **ReLU activation** to introduce non-linearity, enabling the model to learn complex patterns in the data. A **dropout layer** with a 10% rate was applied after each hidden layer to prevent overfitting by randomly dropping units during training.

```
Neural Network Architecture:
ConcreteStrengthANN(
  (fc1): Linear(in_features=7, out_features=128, bias=True)
  (fc2): Linear(in_features=128, out_features=128, bias=True)
  (fc3): Linear(in_features=128, out_features=128, bias=True)
  (dropout): Dropout(p=0.1, inplace=False)
  (fc4): Linear(in_features=128, out_features=1, bias=True)
  (activation1): ReLU()
  (activation2): Identity()
)
```

- **Incorporation of Physics-Based Knowledge**: The key aspect of this model is its physics-informed nature, achieved by incorporating an **empirical equation** that guides the learning process. The empirical equation is:

  "**Concrete Strength = A.ln(age + 1) + C.e$^{(B.\ (UPV\ /\ 1000))}$ + C . ln(ER)**"

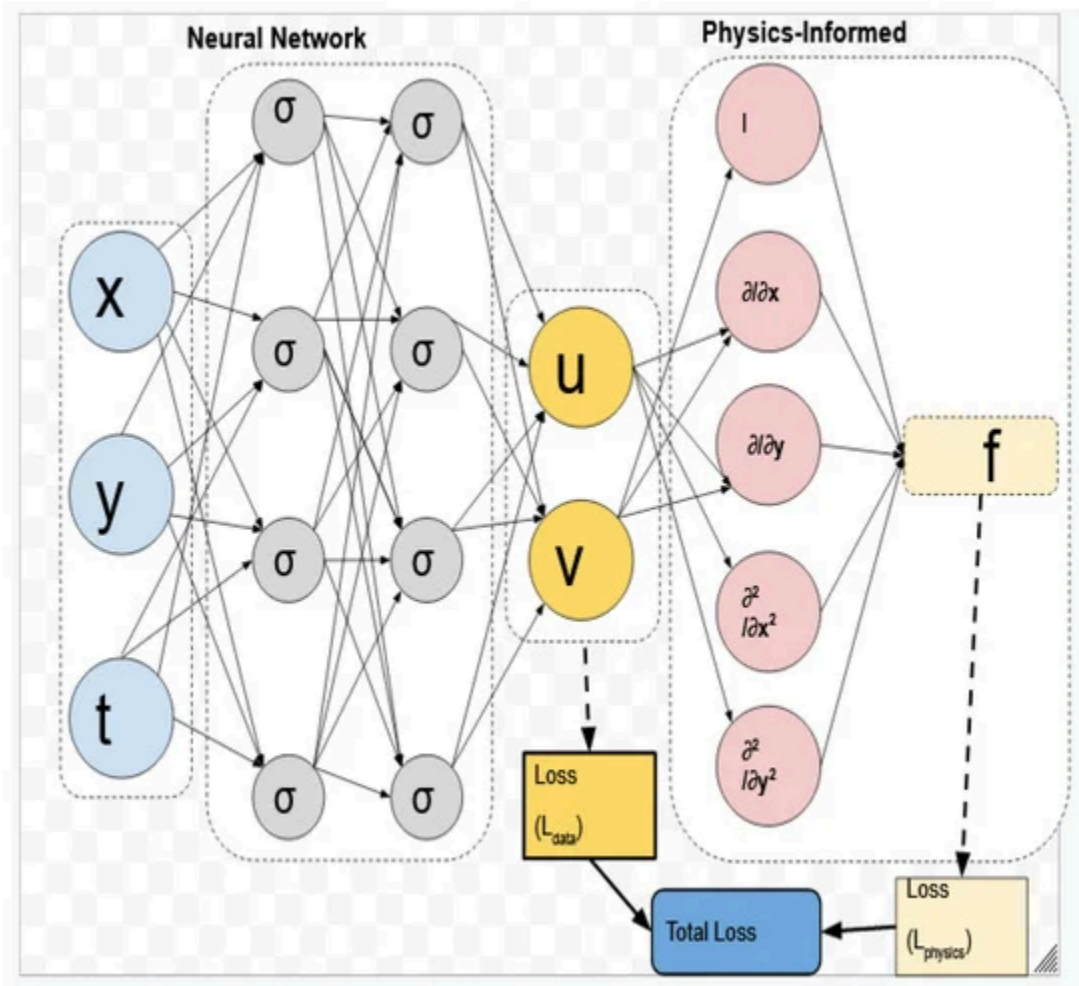
  A, B, Care empirical coefficients derived from existing domain knowledge.

  The equation serves as a physics-based constraint, ensuring that the model learns relationships that are physically meaningful and consistent with known principles.

2.3 Loss Functions and Training Process

- **Loss Function Design**: To effectively train the PINN, we combined multiple loss functions:
  - **Data Loss (Mean Squared Error, MSE)**: This measures the difference between the predicted and actual values of concrete strength, ensuring the model's accuracy in reproducing the data.
  - **Physics-Informed Loss**: This component aims to ensure that the predicted concrete strength values adhere closely to the empirical physical equation. By enforcing this, the model respects known physical relationships even when it might otherwise overfit to noisy data.
  - **Boundary Loss**: The boundary loss adds penalties when predictions violate known physical constraints, such as realistic bounds on curing age, UPV, ER,

and strength limits. This component is critical to ensure the output predictions fall within the physically feasible domain.



- **Total Loss Function**: The overall loss function is a weighted sum of the three components above, defined as:
  "**Total Loss = Data Loss + λ_phy × Physics-Informed Loss + λ_boundary × Boundary Loss**"
  Where **λ_phy** and **λ_boundary** are hyperparameters that control the contributions of physics-based and boundary constraints to the total loss

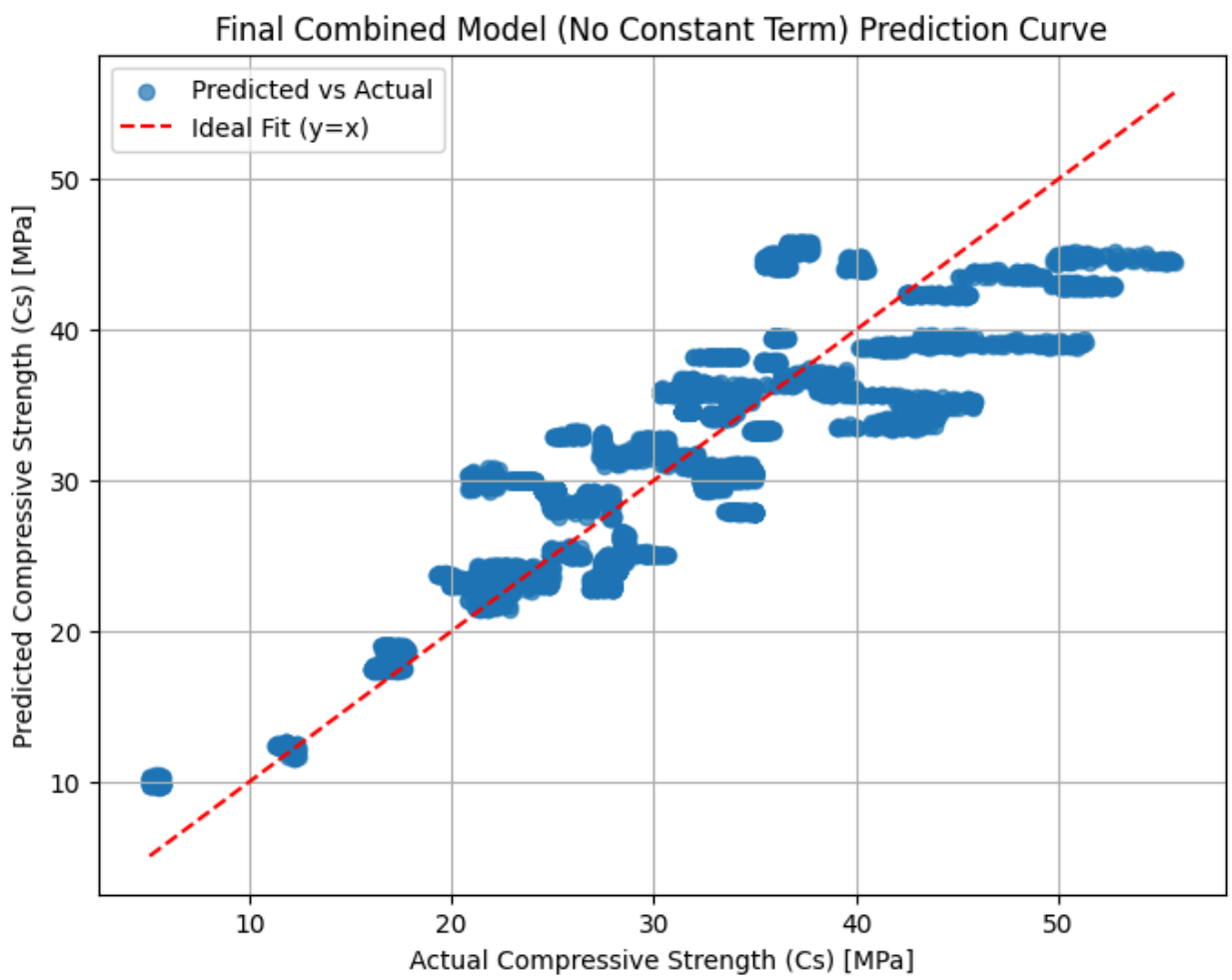
  .

- **Training Procedure**: The model was trained for up to **1000 epochs** using an **early stopping mechanism**. Early stopping was employed to prevent overfitting by halting the training once the validation loss ceased to improve for a certain number of epochs. The **AdamW optimizer** was used with a carefully tuned learning rate to minimize the

loss function, and a **learning rate scheduler** was implemented to reduce the learning rate when the validation performance plateaued. The use of **gradient clipping** ensured stability during the optimization process, particularly in the presence of steep gradients.

3. Insights and Observations

- **Effectiveness of Physics-Informed Neural Networks**: One of the major findings of this project is the effectiveness of integrating empirical equations into the training process. The **PINN** model achieved an **R^2 score of 0.75**, demonstrating a good balance between fitting the training data and adhering to physical laws. This suggests that physics-based constraints can substantially improve generalization, especially in scenarios where data is limited or noisy.

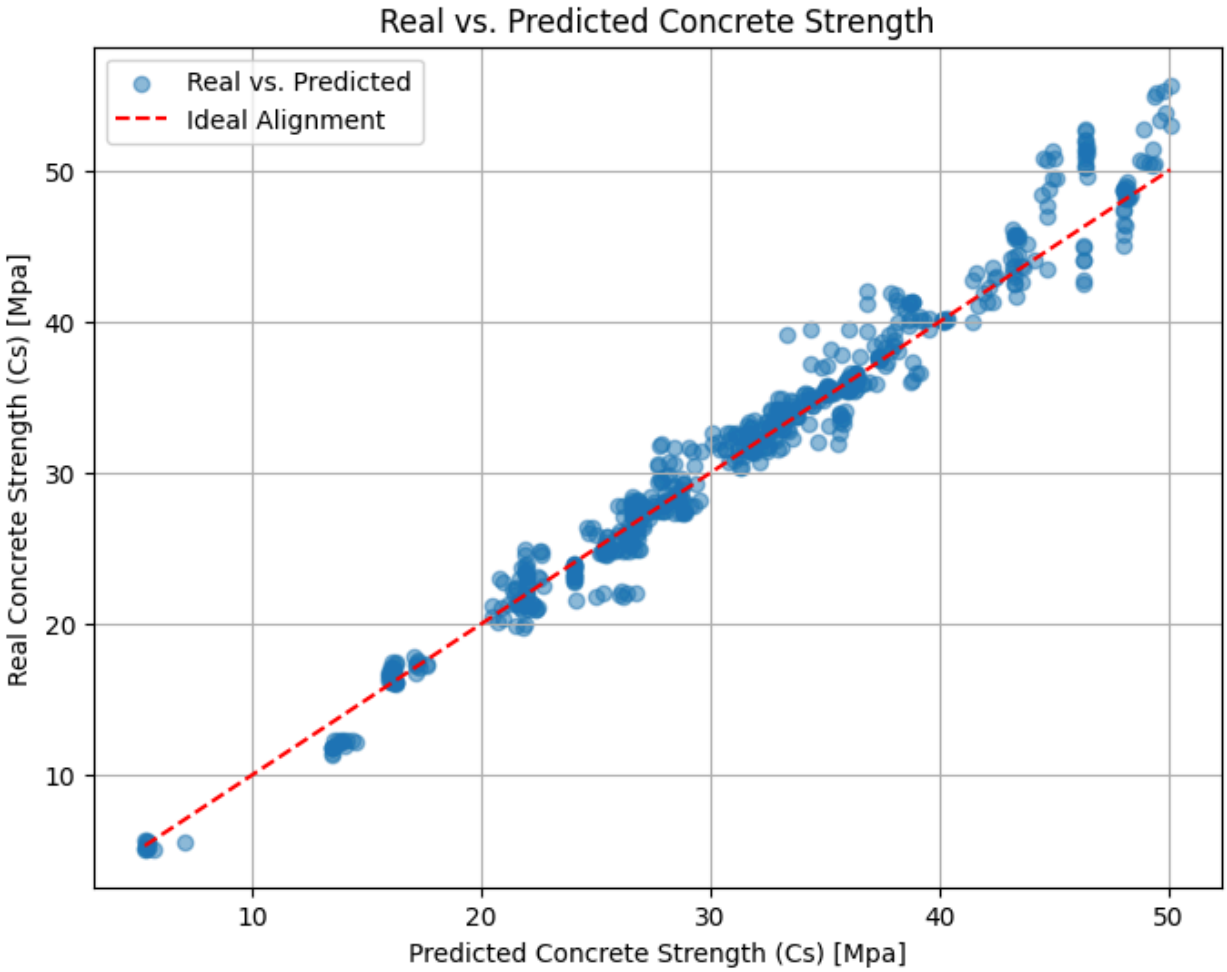


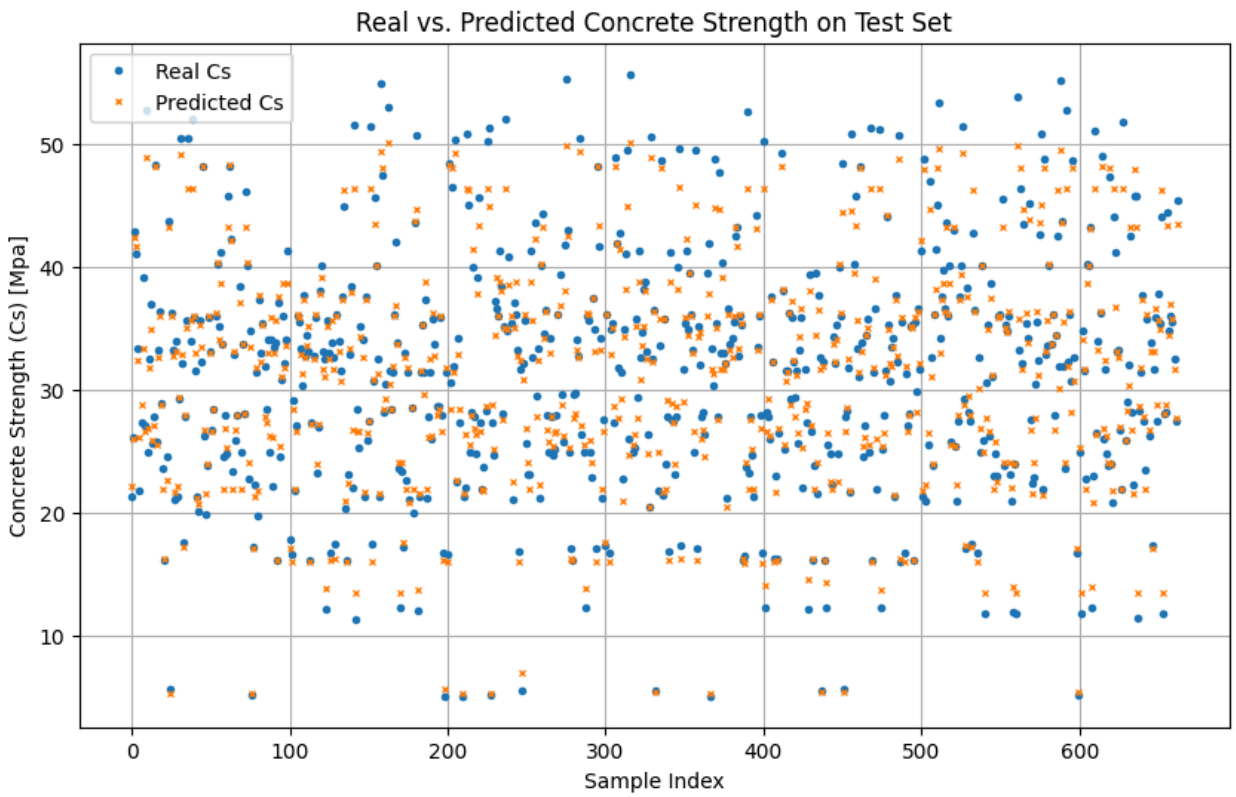
- **Training Dynamics**: The convergence of **training and validation loss curves** was monitored closely throughout the training process. The early stopping mechanism

triggered at around **epoch 457**, indicating that further training would not yield significant improvements. The inclusion of the physics-informed loss helped to stabilize the learning process and ensured that the model predictions remained realistic, thereby avoiding physically implausible predictions.

```
Early stopping at epoch 457
```

- **Visual Analysis**: A detailed **Real Concrete Strength vs. Predicted** scatter plot was generated to evaluate model performance. The majority of the predictions aligned well with the ideal $y = x$ line, indicating that the model effectively captured the relationship between features and concrete strength. However, there were minor discrepancies, likely due to inherent noise in the data, which highlights potential areas for improvement.


Real vs. Predicted Concrete Strength
Real vs. Predicted
Ideal Alignment
Real Concrete Strength (Cs) [Mpa]
Predicted Concrete Strength (Cs) [Mpa]
10
20
30
40
50
10
20
30
40
50



4. RESULT

Test R^2 Score: 0.9723

The **Test R^2 Score** of **0.9723** indicates that the model explains **97.23%** of the variance in the actual concrete strength values from the test set. This is a significant achievement, showing that the Physics-Informed Neural Network (PINN) has effectively learned the underlying patterns in the data and generalized well to unseen data points.

A high R^2 score like **0.9723** means that the model's predictions are very close to the actual values, suggesting that the model is robust and accurate. This result demonstrates the successful integration of empirical physics-based constraints into the neural network, which helped guide the model towards physically realistic predictions while maintaining high accuracy. The remaining **2.77%** of variance, which the model did not explain, could be attributed to factors such as noise in the data or slight imperfections in the empirical relationships modeled.

Overall, the high R^2 score supports the conclusion that the use of physics-informed loss components significantly improved the model's performance, leading to reliable predictions that align well with known physical laws and observed behaviors in concrete strength.

5. Deployment

- **Model Saving and Storage**: After training, the model was saved using PyTorch's `torch.save()` method. Given that the training was performed on **Google Colab**, the trained model was saved directly to **Google Drive** to facilitate easy access for future use and potential re-training.
- **Web Application Integration**: The trained model was integrated into a simple **Flask web application** to provide a user-friendly interface for concrete strength prediction. The web application takes input features such as **Type of Aggregates, Age, UPV, and ER**, and provides a predicted value for concrete strength. This setup makes the model readily usable for practitioners in the construction field to predict concrete strength based on project-specific parameters.

5. Conclusion

This project demonstrates the effectiveness of **Physics-Informed Neural Networks (PINNs)** for predicting concrete strength. By integrating domain knowledge into the learning process, the PINN model is capable of providing predictions that are not only accurate but also physically plausible. This combination of data-driven learning and physical modeling ensures robustness in predictions, even when faced with limited data or noisy inputs. The results indicate that physics-based constraints significantly enhance the reliability of the model, making it a promising approach for practical civil engineering applications.

Future Directions: To further improve the model, future work could focus on enhancing the network's architecture (e.g., experimenting with different activation functions), fine-tuning the loss weights, and expanding the empirical equation to include more factors. The application of PINNs to other aspects of civil engineering, such as predicting structural performance or optimizing construction processes, could also be explored to improve the quality and efficiency of construction projects.

6. Limitations

While the project demonstrated successful predictions of concrete strength using PINNs, there were several limitations that need to be addressed:

- **Data Quality and Quantity**: The dataset used for training was relatively small, which may have limited the model's ability to generalize across a wider range of conditions. Additionally, the data might contain noise or errors that could impact model accuracy.
- **Empirical Equation Limitations**: The empirical equation used as the physics-based constraint may not fully capture all the complexities involved in concrete strength prediction. Concrete behavior is influenced by numerous factors, and the current equation may lack certain critical parameters that could improve the model’s precision.
- **Computational Cost**: Training the PINN model was computationally expensive due to the combination of data-driven and physics-informed losses. The need to compute

additional physics-informed and boundary losses led to increased training time, which could be a limitation for real-time applications or larger datasets.

- **Hyperparameter Sensitivity**: The model's performance was sensitive to the choice of hyperparameters, such as the weights for the loss components (λ_phy and λ_boundary). Tuning these hyperparameters required a significant amount of trial and error, which can be time-consuming and may not guarantee optimal results.
- **Assumptions in Boundary Conditions**: The boundary conditions incorporated into the model were simplified to cover typical ranges for curing age, UPV, and ER. However, in practical scenarios, concrete properties can vary widely, and the simplified assumptions might limit the model's applicability in certain edge cases.
- **Limited Real-World Testing**: The model has not been extensively tested in real-world scenarios or across different types of concrete mixtures. The lack of practical validation means that while the model performs well on the test set, its robustness in diverse conditions remains uncertain.

References:

- Kazemifard, S., Motaghed, S. & Eftekhari, N. NDT prediction of self-compacting concrete strength based on maturity method. *Multiscale and Multidiscip. Model. Exp. and Des.* **7**, 1031–1043 (2024).

- Correlation between concrete strength properties and surface electrical resistivity
  Correlação entre as propriedades de resistência do concreto e a resistividade elétrica superficial

- Compressive strength prediction of Portland cement concrete with age using a new model
- **Physics-Informed Neural Network (PINN) Evolution and Beyond: A Systematic Literature Review and Bibliometric Analysis**